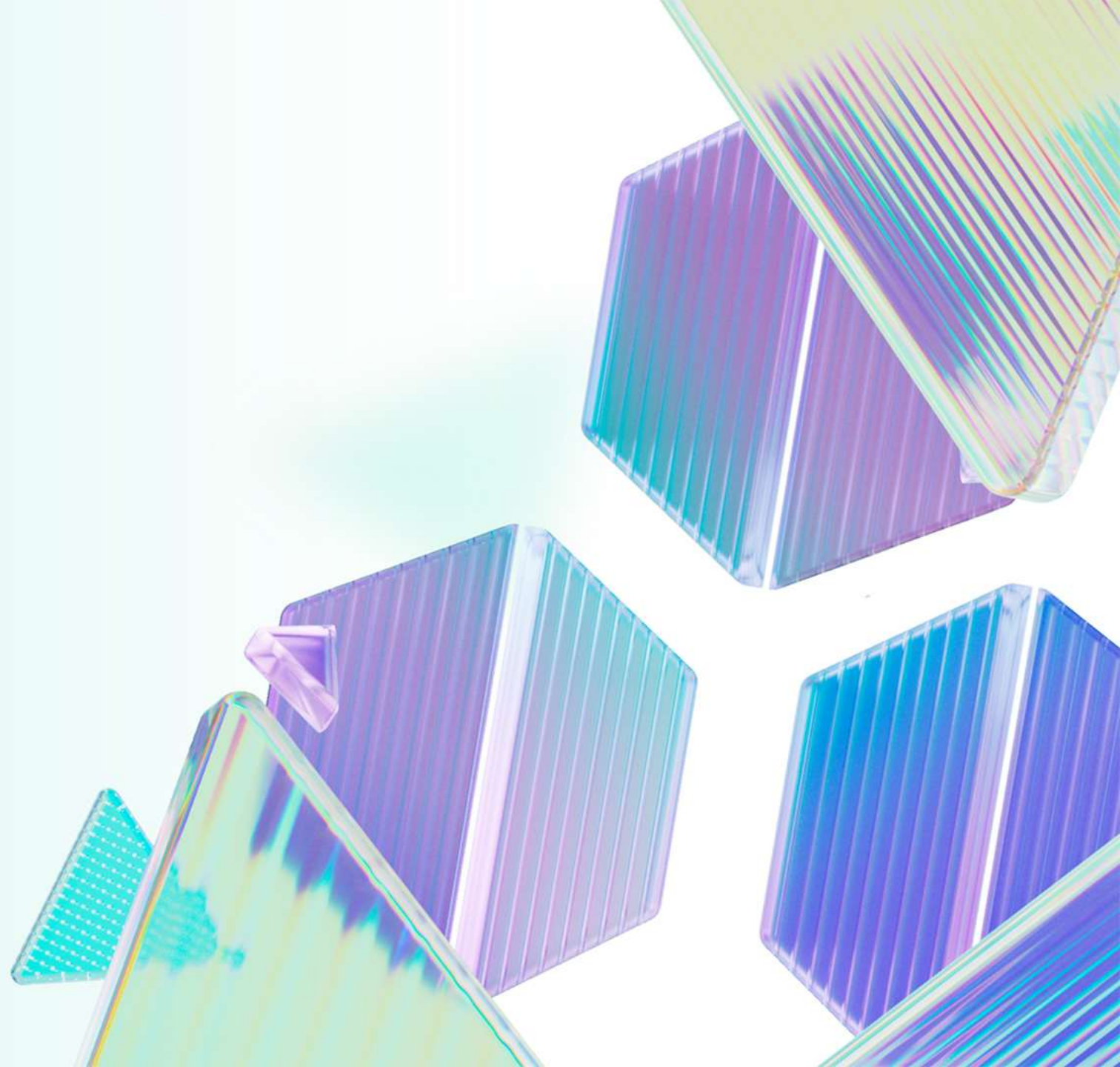




Fabric security: everything you need to know!

Vengatesh Parasuraman

Principal Program Manager,
Microsoft



For all the latest security related
information please visit:

aka.ms/MSFabricSecurity

This slide deck:



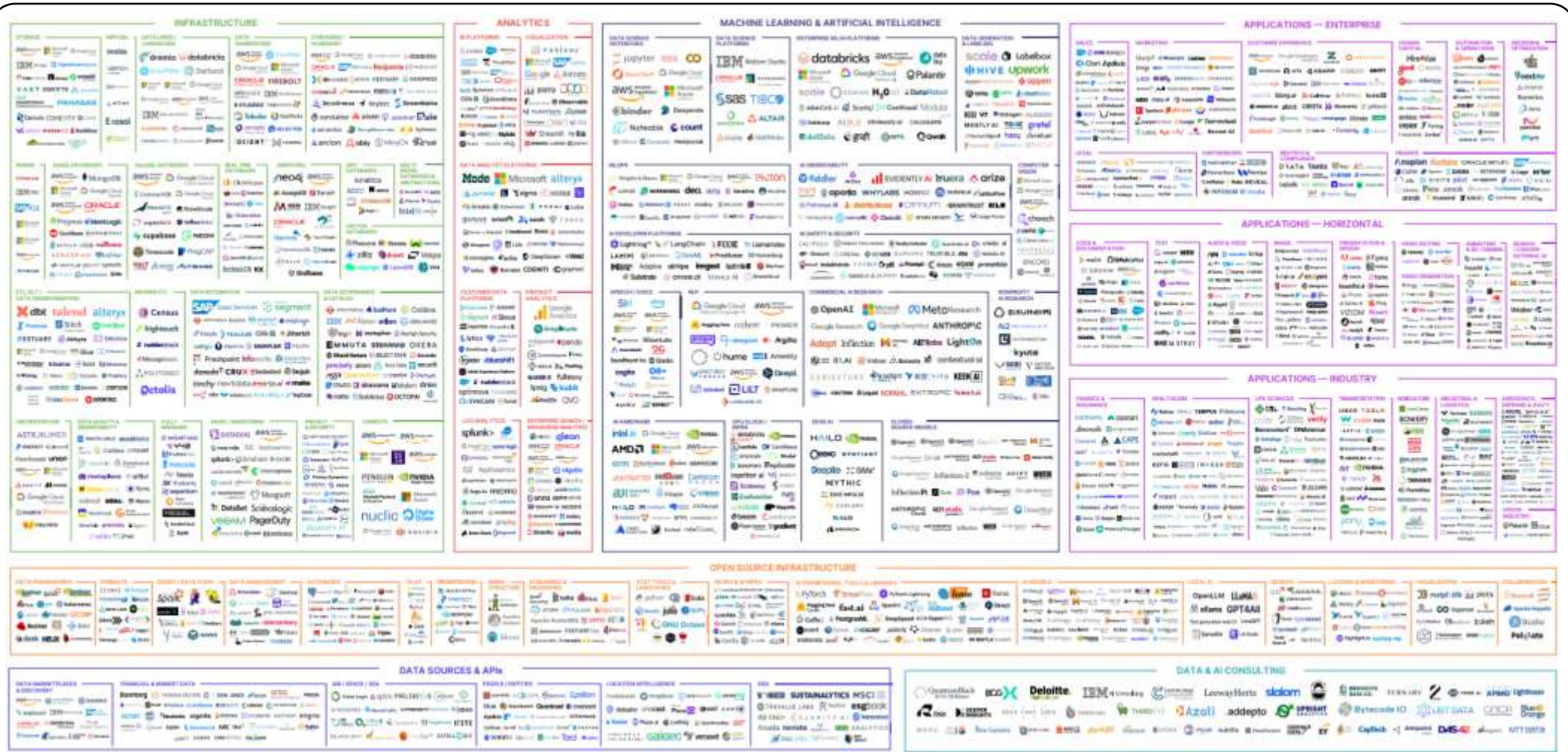
Agenda

- 1 Intro to Fabric as a SaaS platform
- 2 Network and Identity security
- 3 Workspace & item security
- 4 Secure and compliant by default
- 5 Data security & governance

A world awash with data...

**How do you translate data into
competitive advantage?**

The 2024 ML, AI, and Data Landscape



“

Simplify,
I am the Chief Data Officer
and don't want to be the Chief Integration Officer.

Every CDO, Every Enterprise

”



Microsoft Fabric

Data analytics for the era of AI

Complete Analytics Platform

Unified product, experience,
and architecture

Delivered as SaaS



Microsoft Fabric

Data analytics for the era of AI

Complete Analytics Platform

Unified product, experience,
and architecture

Delivered as SaaS

Lake Centric and Open

Common SaaS data lake shared
by all compute engines

Deep commitment
for open formats and APIs



Microsoft Fabric

The unified data platform for the era of AI



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI



AI



OneLake



Purview

Microsoft Fabric

End-to-end analytics data fabric From the data lake to the business user

Complete Analytics Platform

- Everything, Best of Breed
- Unified
- Azure OpenAI Copilots

Lake centric and open

- OneLake
- One Copy
- Always Synced

Empower Every Office User

- Familiar and intuitive
- Built into Office
- Insight to action

Pervasive security and governance

- End to end visibility
- Always governed
- Secure by default

Microsoft Fabric

Pervasive Security and Governance for the cloud world

Complete Analytics Platform

- Everything, Best of Breed
- Unified
- Azure OpenAI Copilots

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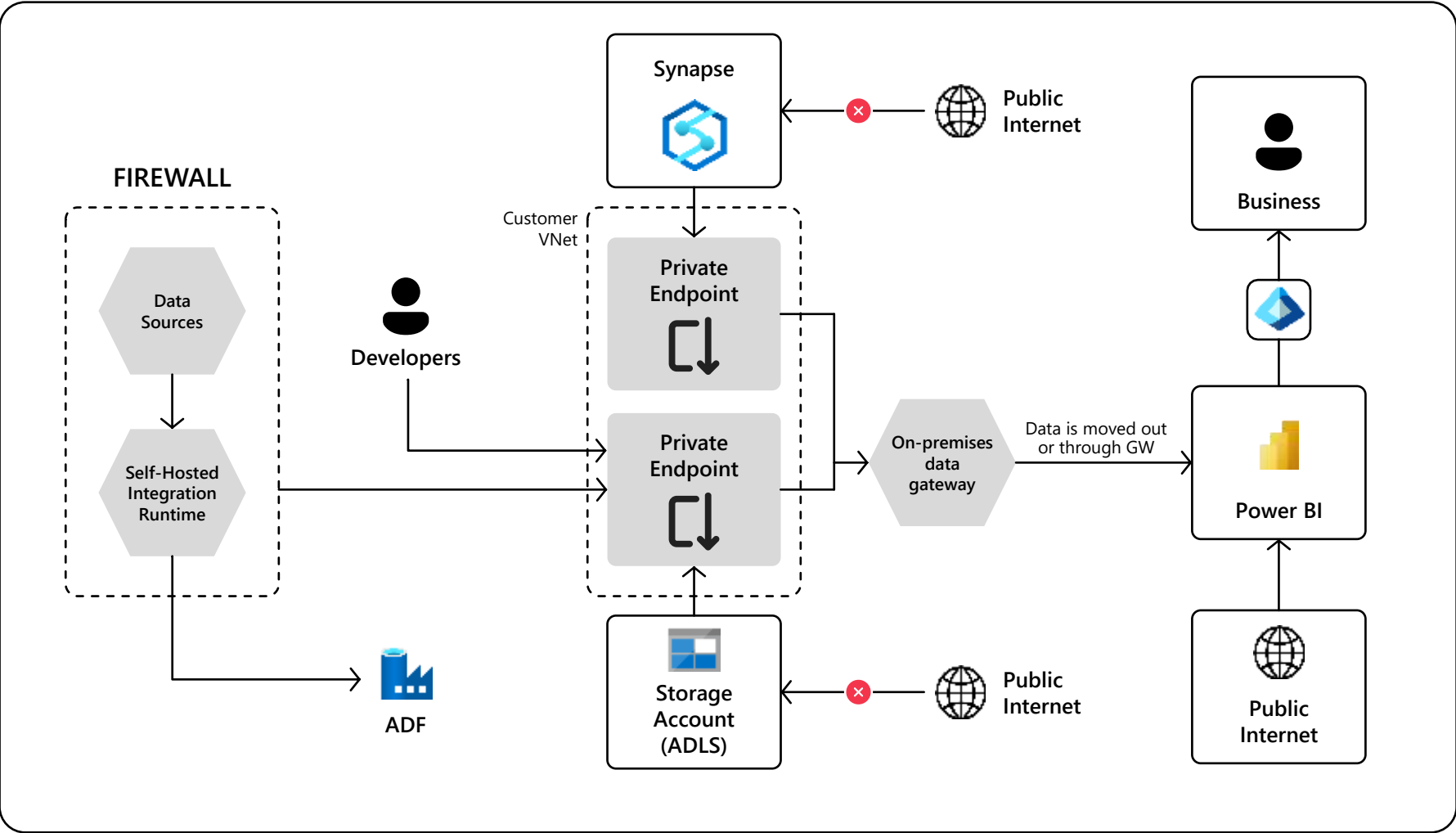
Modern data challenges

- Bring data to the masses, everyone should have access to data to make decisions
- Instant access to the latest data, no copying around
- Modern workforce, anywhere, any device
- At the same time, you need to secure, govern and audit your data to protect customers and the company
- SaaS platforms are designed with these challenges in mind.
- Shift from siloed PaaS to integrated SaaS

Common network security requirements

- ➔ Need to be able to connect to data inside a firewall\private link from Fabric (outbound)
- ➔ Inbound protection (restrict inbound by network location)
- ➔ Secure access to the “backend services”
- ➔ More stringent customers (FSI\HLS):
 - Traffic needs to be private (not via public internet)
 - Endpoints should not be open to the public internet

Existing PaaS World



PaaS services are disconnected from the Public internet.

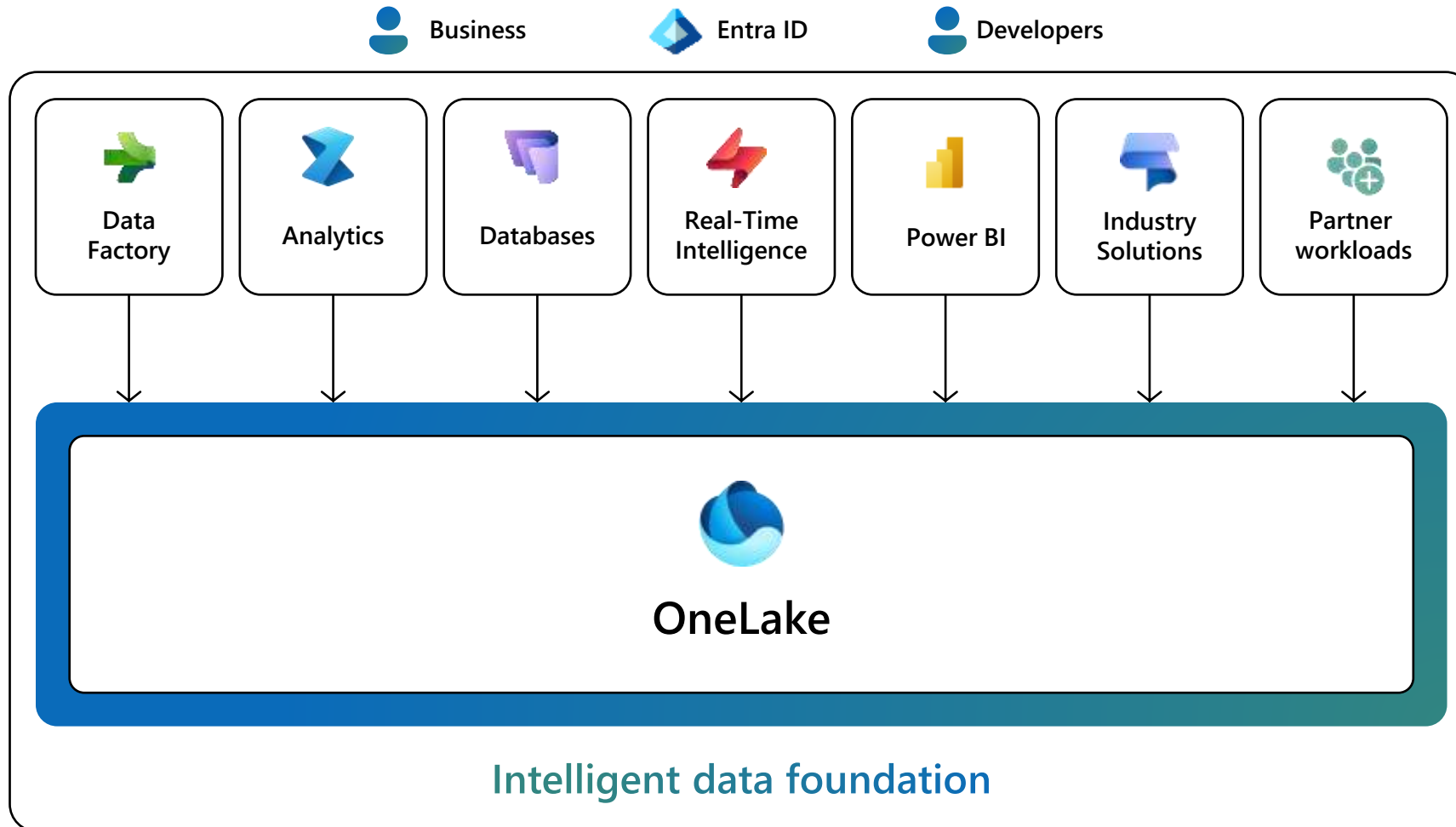
All communications between the services are configured to go through private endpoint

Developers connect to a VPN and connect to the backend services

Data sources are connected through the self-hosted runtime inside the firewall

When the CEO access the data in Power BI Report, the data is imported into Power BI or through the GW with DQ.

Microsoft Fabric – SaaS World



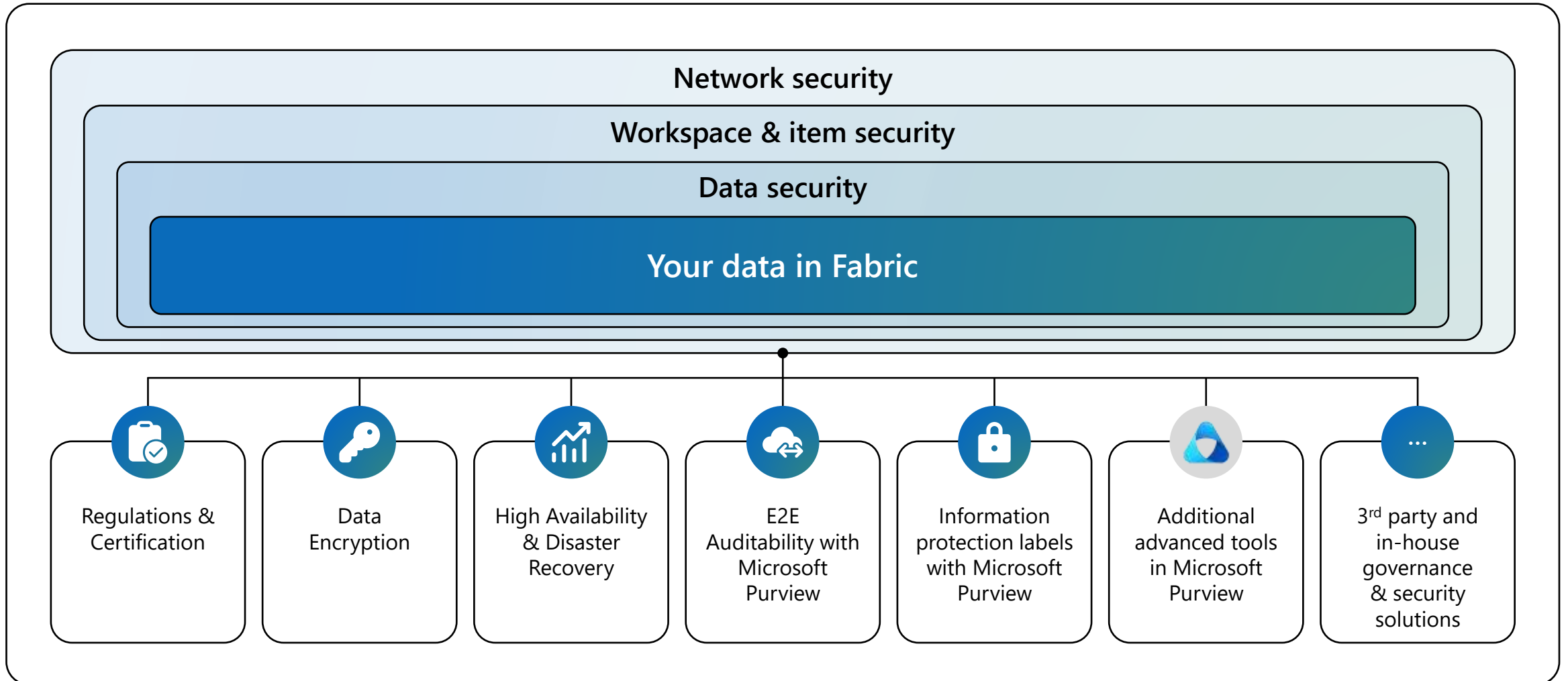
Users connect to the SaaS service from global network

Endpoints and access are protected using Entra ID

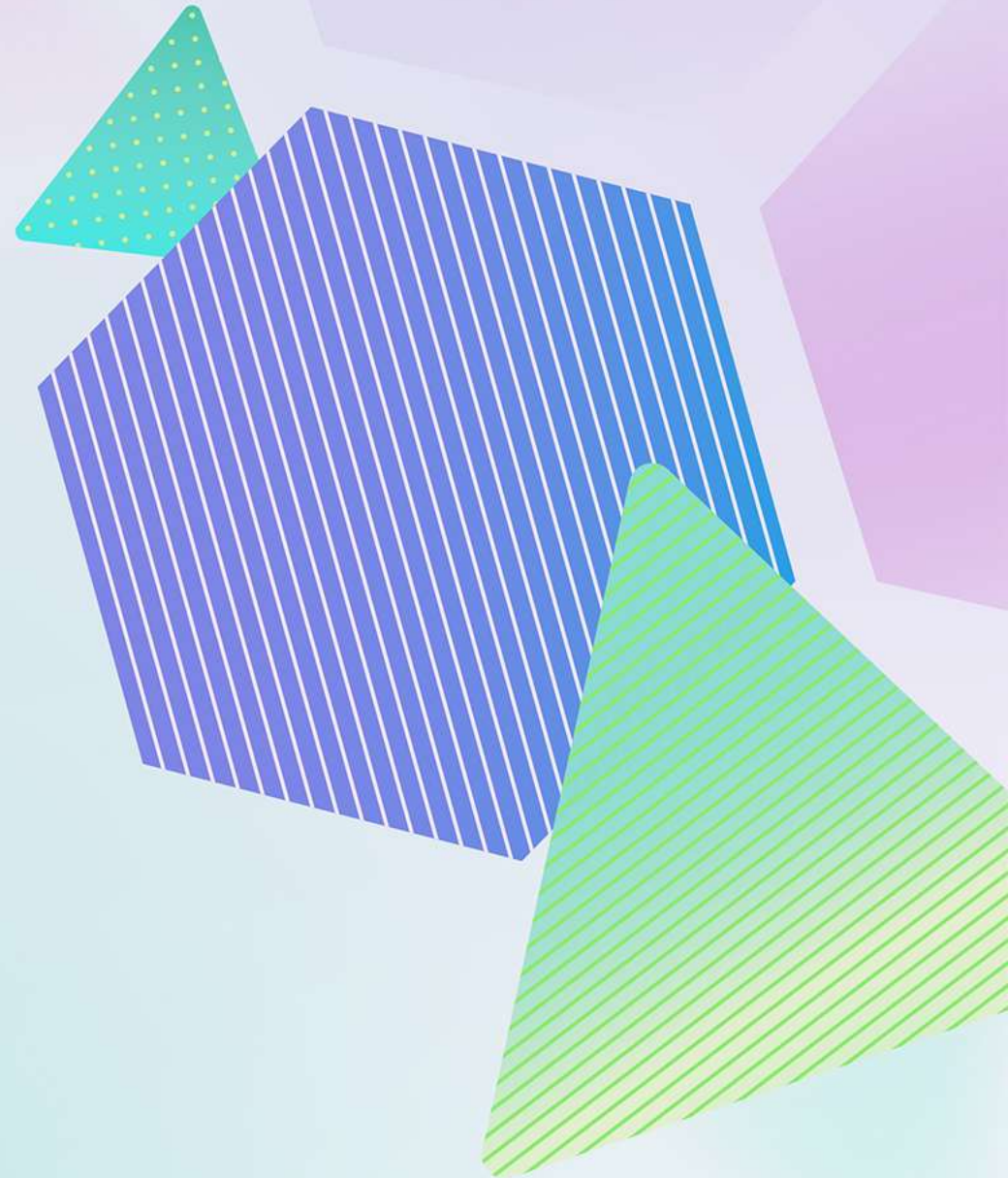
All internal communications between the experiences happens through MS backbone network

When the CEO access the data in Power BI Report, the data is fetched directly from the OneLake instantly and securely without copying or moving

Security layers in Microsoft Fabric

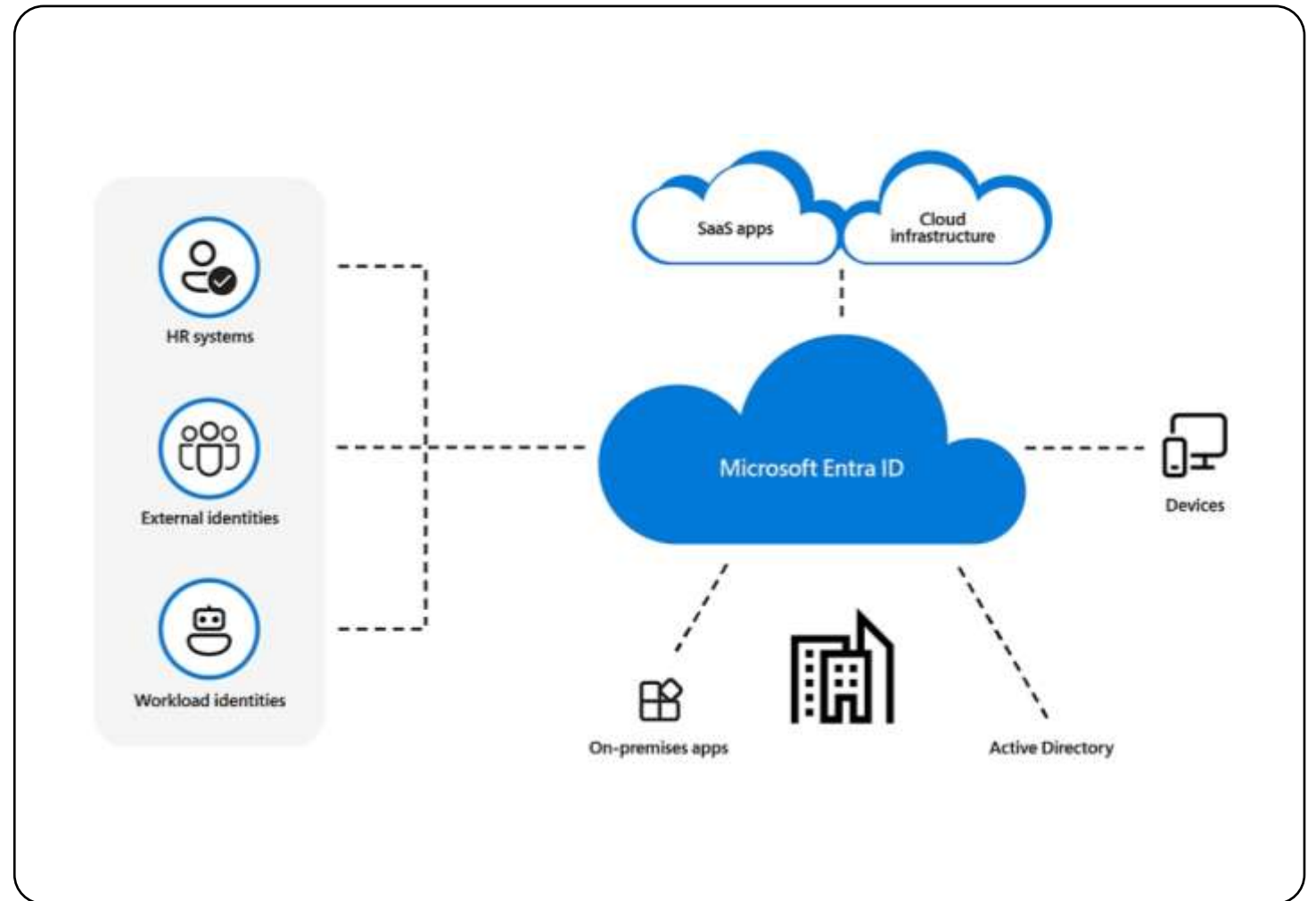


Network security



Authentication

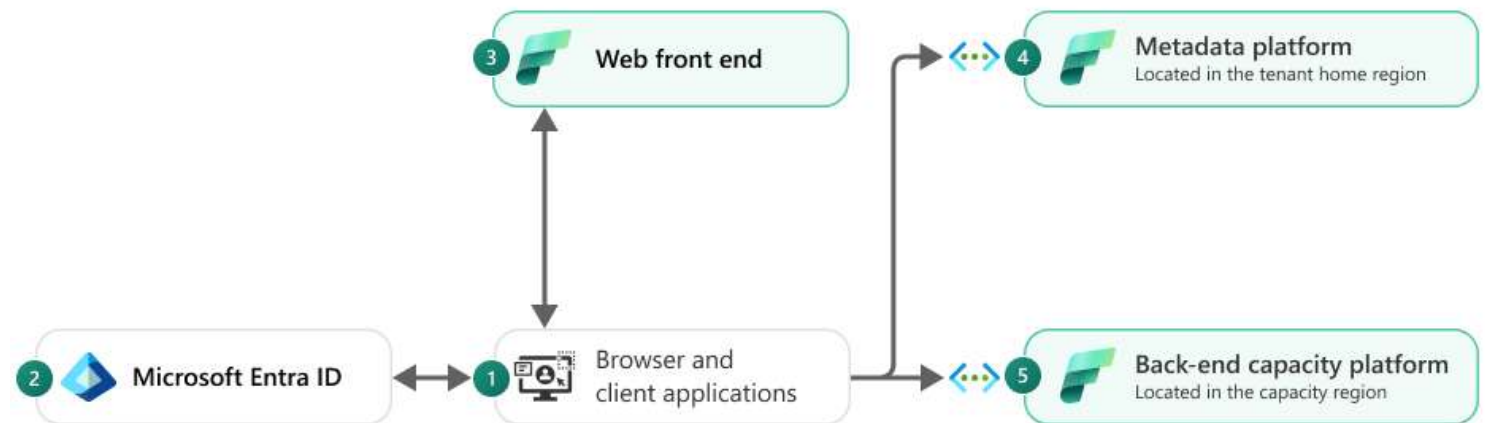
Microsoft Entra ID at
the hearth of your security



Microsoft Fabric Architecture

Fabrics is built as a SaaS product

- Users connect to "Front End Services" and we are using "Entra ID" to authenticate and **trust all requests**.
- All the clusters are **protected behind "back-end services" and v-nets**.
- Traffic between experiences is **going over MS backend network**
- Traffic to Fabric will be using **at least TLS 1.2**



Inbound protection options

Perimeter Network Security



From limited known locations

Zero Trust Approach



to unknown locations

New principles for Zero Trust in today's reality



Verify explicitly



Use least privilege access



Assume breach

Fabric leverages Entra ID for secure adaptive access

Protect access to resources and data with strong authentication and risk-based access policies



User-friendly multifactor authentication (MFA) and Passwordless authentication support



Configurable Conditional Access policies based on context and risk assessment ¹

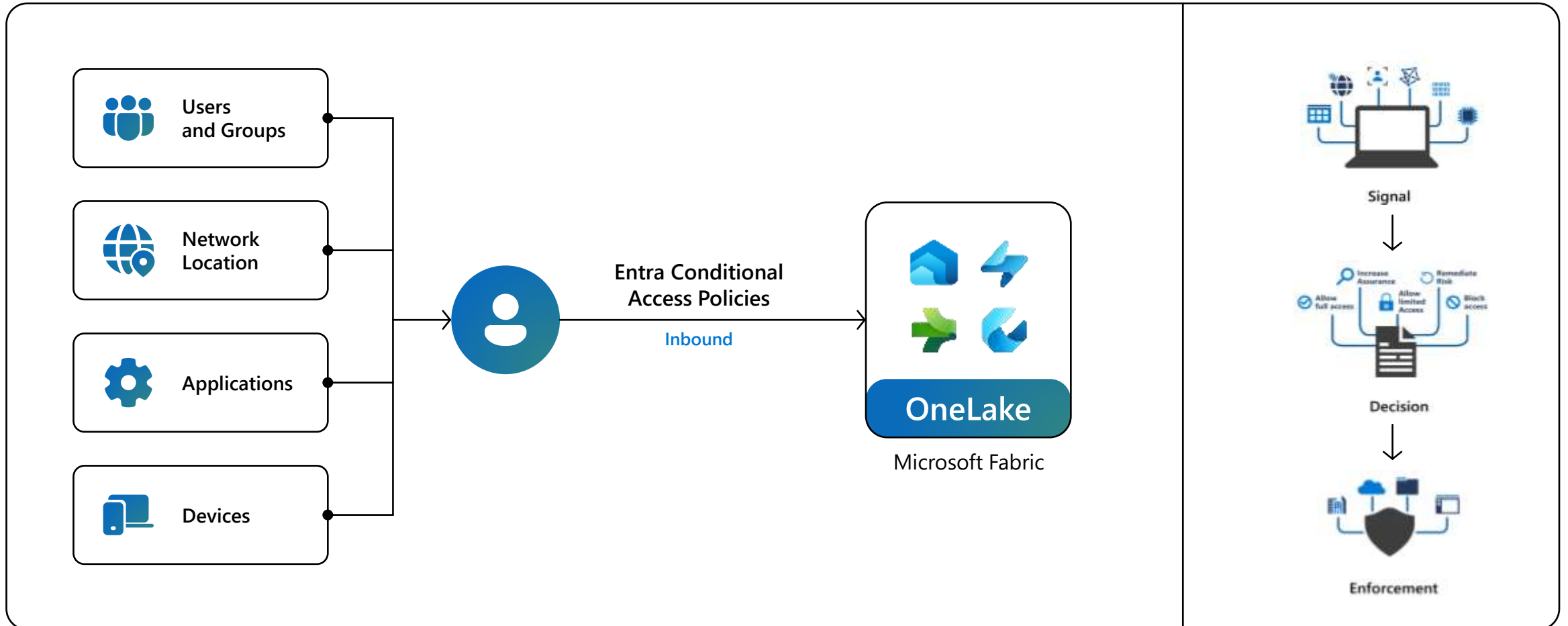


User and entity behavior analytics (UEBA) to automatically protect against identity compromise ²

¹Microsoft Entra ID P1 \ E3 license required  ²Microsoft Entra ID P2 \ E5 license required

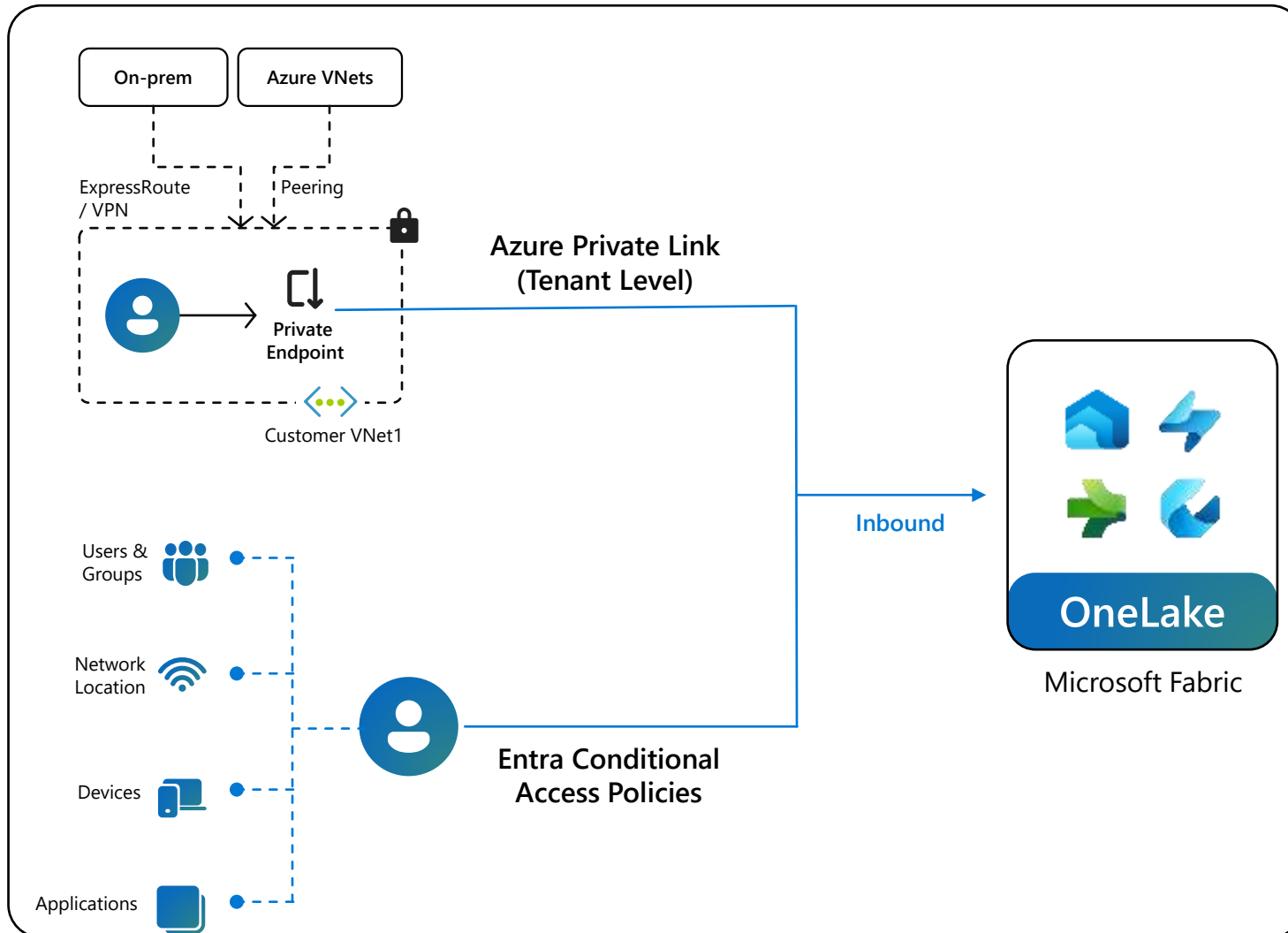
Conditional Access Policies

Common Decisions – Block, Grant, Require MFA



Private Link for Fabric

Perimeter Network Security for your tenant

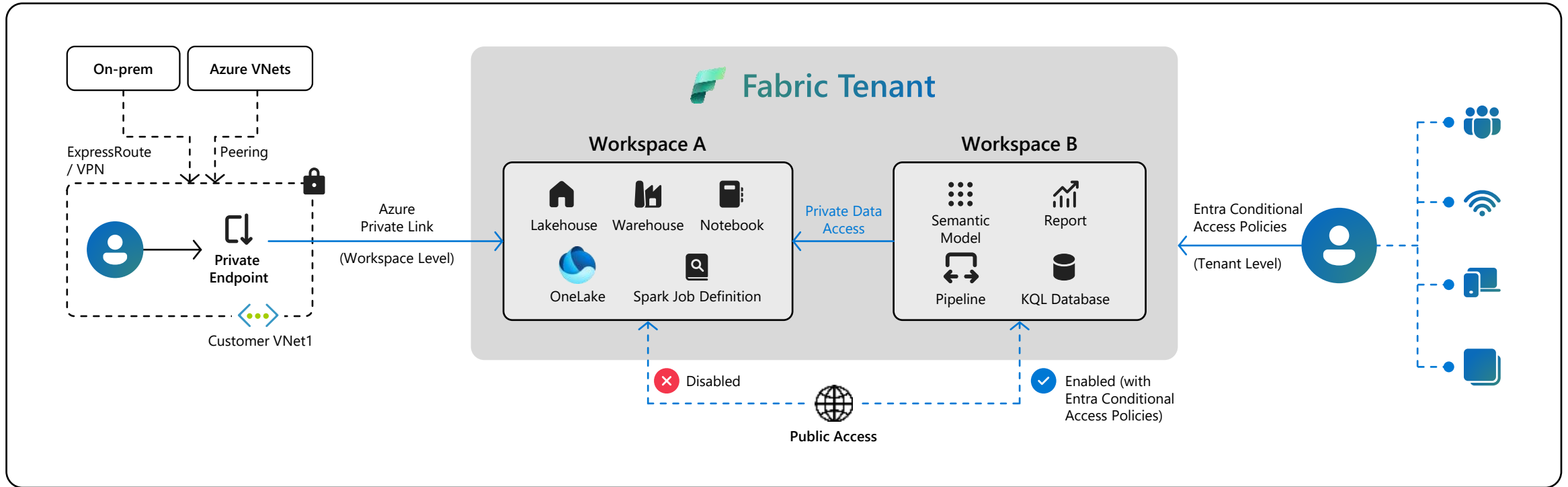


What it means:

1. Fabric is disconnected from the public internet
2. Every users needs to connect to the private network to get access on every device
3. No longer able to load resources locally (slower reports)
4. Increases ExpressRoute bandwidth and added costs for Private Links
5. Several product limitations (like on-prem data gateway)

Workspace level Private Link for Fabric

Perimeter Network Security for your workspace



Selected workspaces can be protected using Private Links and closed from public internet.

Create a secure connection between public and private workspaces using private data access.

Public workspaces are secured through Entra policies for example to use Power BI.

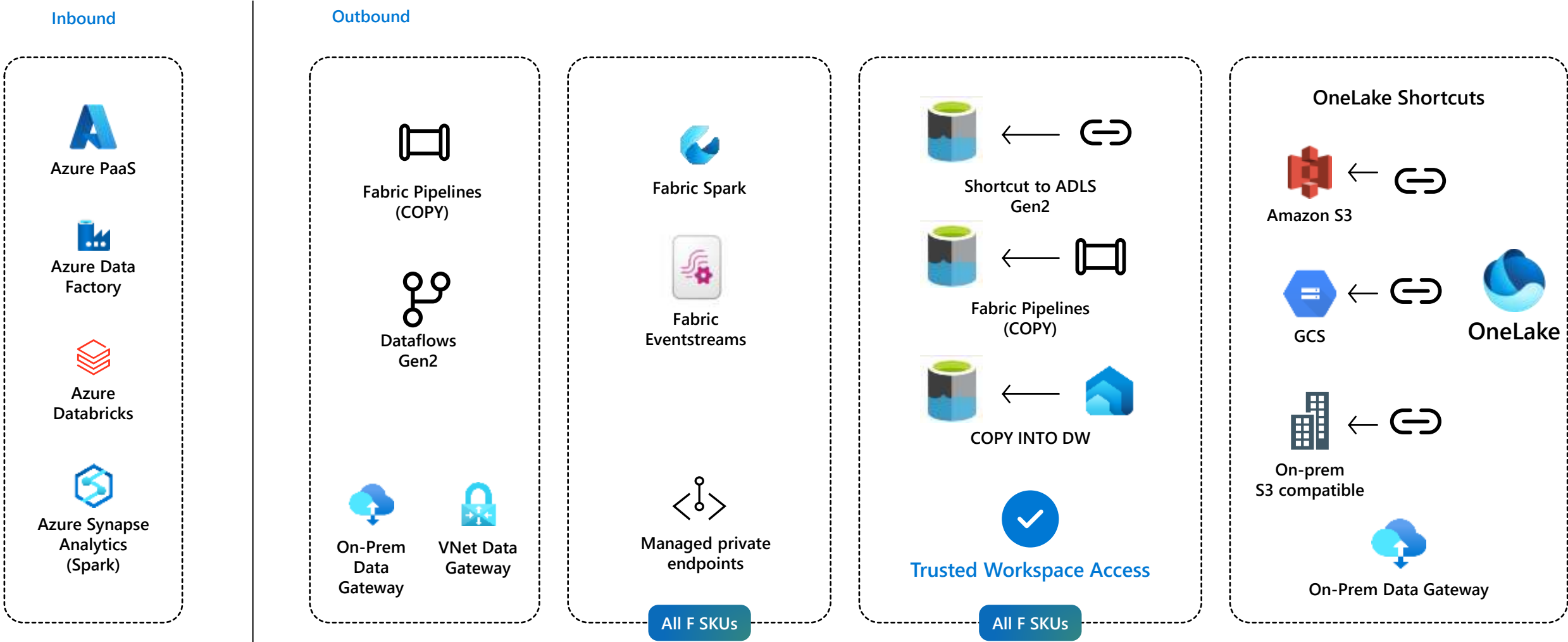
Limitations of tenant wide PL don't apply !!!

Getting data into Fabric

How to connect/load data in the private network from Fabric

Inbound

Outbound



Workspace outbound access protection

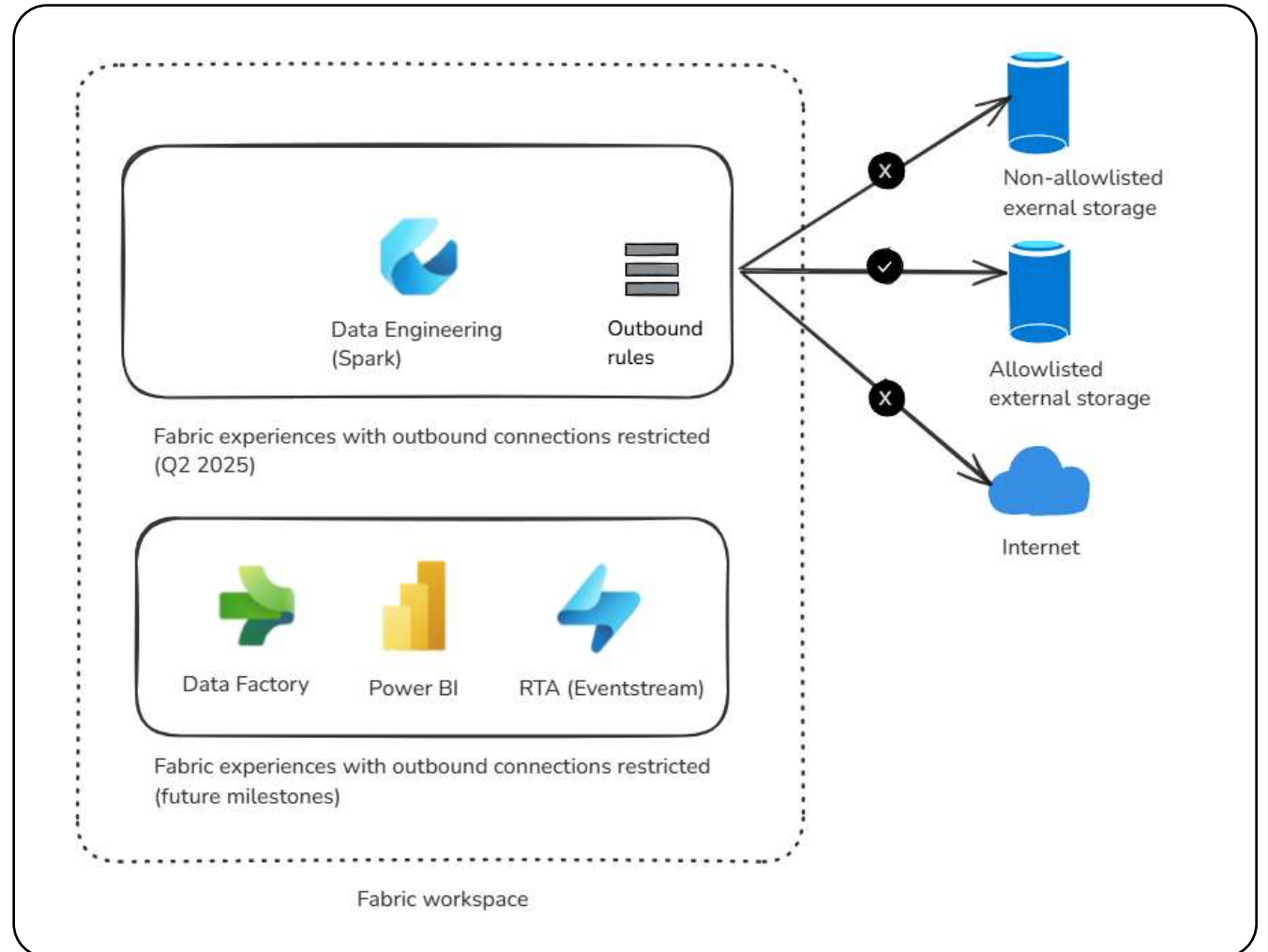
Workspace outbound access protection

Workspace level control to restrict outbound access to Public network

Restrict outbound connections to permitted destinations only

Use in conjunction with other security features to achieve robust outbound control

Upcoming: Q2 2025 for Spark. Other workloads in future milestones



Workspace outbound access monitoring

Fabric allows tracking user activities for connection creation and use in usage logs and purview audit.

For example:

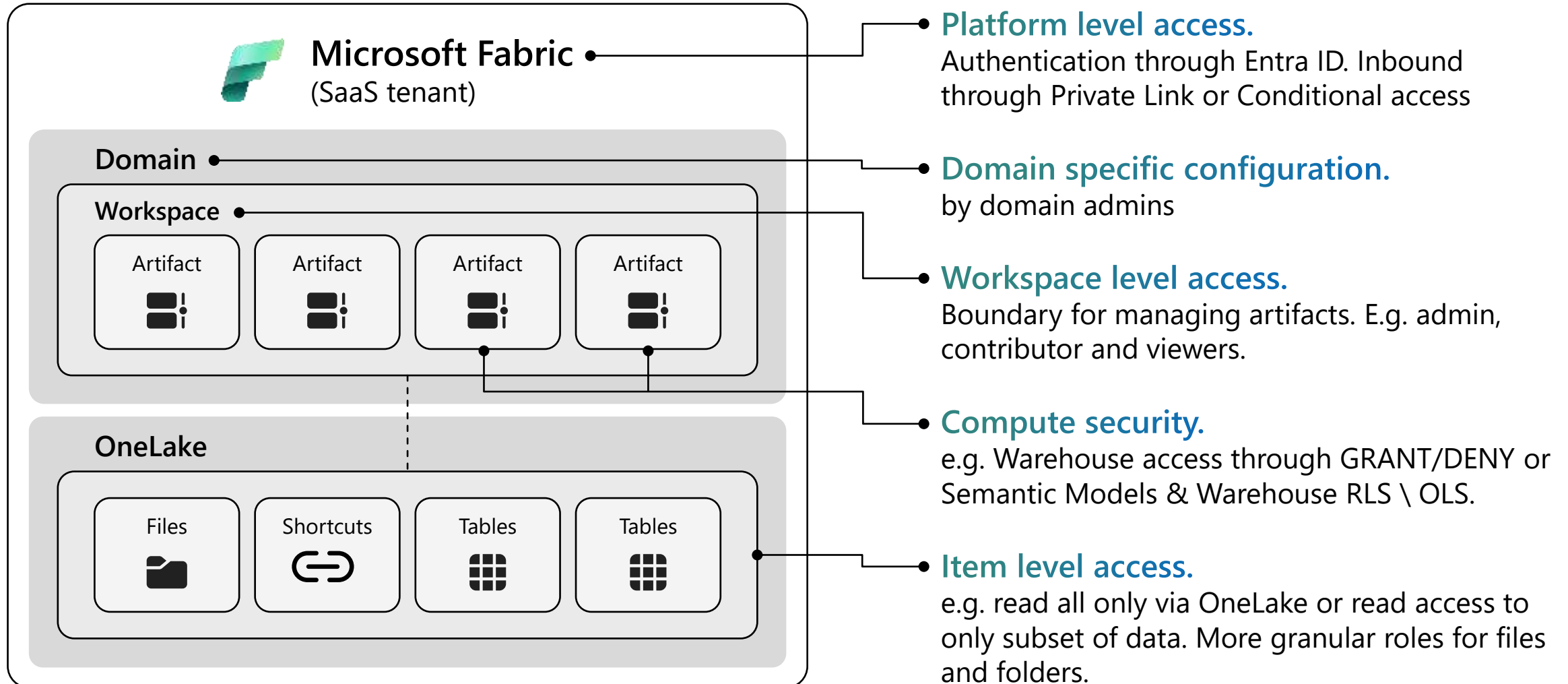
- Create cloud connections
- Create notebooks \ pipelines
- Create shortcuts

For more details:

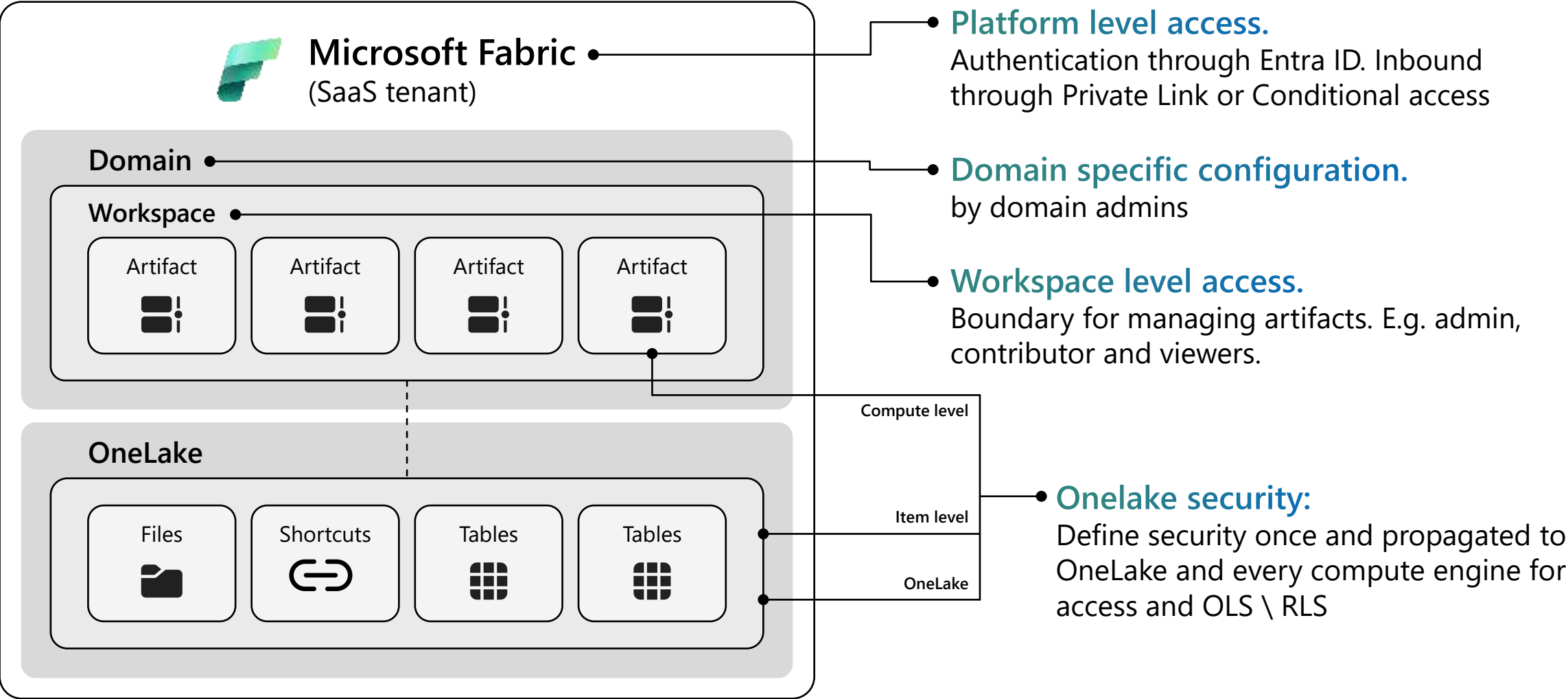
- [Track user activities in Microsoft Fabric - Microsoft Fabric | Microsoft Learn](#)
- [Data connection auditing for exfiltration protection | Microsoft Learn](#)

Workspace & item security

Multiple layers of security and access control



OneLake security public preview soon





OneLake security in Fabric



Secure once and enforce across engines

Define security once in OneLake, and all engines will enforce it uniformly as they process queries and jobs, streamlining data security management and optimizing processes.



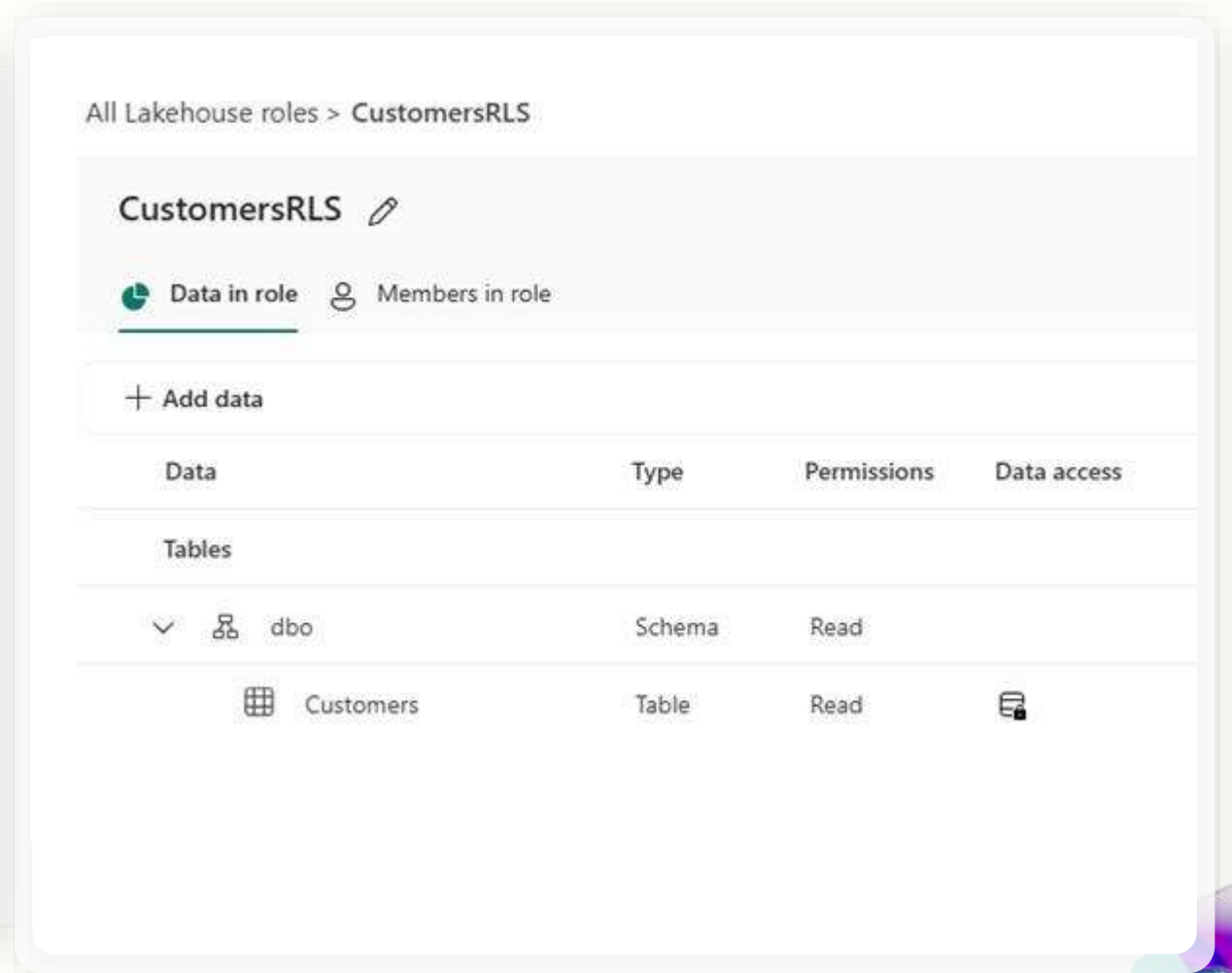
Define granular security for secure sharing

Create roles within a data item that grants access only to granular items with a lakehouse. This includes tables and folders or rows (RLS) and columns (CLS)



Centrally manage all permissions in Fabric

Use OneLake security to centrally manage roles and their permissions across all domains and workspaces in Fabric.



Home

Home

Copilot

Create

Browse

OneLake

Apps

Metrics

Monitor

Learn

Real-Time

Workloads

Workspaces

OneSecurity Test

NorthwindTest

NorthwindTest

...

Power BI

Get data

New semantic model

Open notebook

Manage OneLake data access (preview)

Explorer

Search schemas and tables

NorthwindTest

Tables

dbo

Customers

Employees

EmployeeTerritor...

OrderDetails

Orders

Products

Regions

Shippers

ShortcutOrders

Suppliers

Territories

Files

Orders

Showing 830 rows

	orderID	customerID	employeeID	orderDate	requiredDate	shippedDate	shipVia	freight	shipName	shipAddress	shipCity	shipRegion	shipPostalC...
1	10257	HILAA	4	7/16/1996 12:00:00	8/13/1996 12:00:00	1996-07-22 00:00:00	3	81.91	HILARION-Abast...	Carrera 22 con A...	San Cristóbal	Táchira	5022
2	10262	RATTC	8	7/22/1996 12:00:00	8/19/1996 12:00:00	1996-07-25 00:00:00	3	48.29	Rattlesnake Can...	2817 Milton Dr.	Albuquerque	NM	87110
3	10268	GROSR	8	7/30/1996 12:00:00	8/27/1996 12:00:00	1996-08-02 00:00:00	3	66.29	GROSELLA-Rest...	5ª Ave. Los Palos...	Caracas	DF	1081
4	10269	WHITC	5	7/31/1996 12:00:00	8/14/1996 12:00:00	1996-08-09 00:00:00	1	4.56	White Clover Ma...	1029 - 12th Ave...	Seattle	WA	98124
5	10283	LILAS	3	8/16/1996 12:00:00	9/13/1996 12:00:00	1996-08-23 00:00:00	3	84.81	LILA-Supermerc...	Carrera 52 con A...	Barquisimeto	Lara	3508
6	10287	RICAR	8	8/22/1996 12:00:00	9/19/1996 12:00:00	1996-08-28 00:00:00	3	12.76	Ricardo Adocica...	Av. Copacabana ...	Rio de Janeiro	RJ	02389-890
7	10290	COMMI	8	8/27/1996 12:00:00	9/24/1996 12:00:00	1996-09-03 00:00:00	1	79.7	Comércio Mineiro	Av. dos Lusíadas...	Sao Paulo	SP	05432-043
8	10296	LILAS	6	9/3/1996 12:00:00	10/1/1996 12:00:00	1996-09-11 00:00:00	1	0.12	LILA-Supermerc...	Carrera 52 con A...	Barquisimeto	Lara	3508
9	10305	OLDWO	8	9/13/1996 12:00:00	10/11/1996 12:00:00	1996-10-09 00:00:00	3	257.62	Old World Delic...	2743 Bering St.	Anchorage	AK	99508
10	10309	HUNGO	3	9/19/1996 12:00:00	10/17/1996 12:00:00	1996-10-23 00:00:00	1	47.3	Hungry Owl All-...	8 Johnstown Road	Cork	Co. Cork	NULL
11	10316	RATTC	1	9/27/1996 12:00:00	10/25/1996 12:00:00	1996-10-08 00:00:00	3	150.15	Rattlesnake Can...	2817 Milton Dr.	Albuquerque	NM	87110
12	10317	LONEP	6	9/30/1996 12:00:00	10/28/1996 12:00:00	1996-10-10 00:00:00	1	12.69	Lonesome Pine ...	89 Chiaroscuro ...	Portland	OR	97219
13	10324	SAVEA	9	10/8/1996 12:00:00	11/5/1996 12:00:00	1996-10-10 00:00:00	1	214.27	Save-a-lot Mark...	187 Suffolk Ln.	Boise	ID	83720
14	10330	LILAS	3	10/16/1996 12:00:00	11/13/1996 12:00:00	1996-10-28 00:00:00	1	12.75	LILA-Supermerc...	Carrera 52 con A...	Barquisimeto	Lara	3508
15	10338	OLDWO	4	10/25/1996 12:00:00	11/22/1996 12:00:00	1996-10-29 00:00:00	3	84.21	Old World Delic...	2743 Bering St.	Anchorage	AK	99508
16	10346	RATTC	3	11/5/1996 12:00:00	12/17/1996 12:00:00	1996-11-08 00:00:00	3	142.08	Rattlesnake Can...	2817 Milton Dr.	Albuquerque	NM	87110
17	10347	FAMIA	4	11/6/1996 12:00:00	12/4/1996 12:00:00	1996-11-08 00:00:00	3	3.1	Familia Arquibal...	Rua Orós 92	Sao Paulo	SP	05442-030
18	10349	SPLIR	7	11/8/1996 12:00:00	12/6/1996 12:00:00	1996-11-15 00:00:00	1	8.63	Split Rail Beer & ...	P.O. Box 555	Lander	WY	82520
19	10355	AROUT	6	11/15/1996 12:00:00	12/13/1996 12:00:00	1996-11-20 00:00:00	1	41.95	Around the Horn	Brook Farm Strat...	Colchester	Essex	CO7 6JX
20	10357	LILAS	1	11/19/1996 12:00:00	12/17/1996 12:00:00	1996-12-02 00:00:00	3	34.88	LILA-Supermerc...	Carrera 52 con A...	Barquisimeto	Lara	3508
21	10373	HUNGO	4	12/5/1996 12:00:00	1/2/1997 12:00:00	1996-12-11 00:00:00	3	124.12	Hungry Owl All-...	8 Johnstown Road	Cork	Co. Cork	NULL
22	10379	QUEDE	2	12/11/1996 12:00:00	1/8/1997 12:00:00	1996-12-13 00:00:00	1	45.03	Que Delícia	Rua da Panificad...	Rio de Janeiro	RJ	02389-673
23	10380	HUNGO	8	12/12/1996 12:00:00	1/9/1997 12:00:00	1997-01-16 00:00:00	3	35.03	Hungry Owl All-...	8 Johnstown Road	Cork	Co. Cork	NULL
24	10381	LILAS	3	12/12/1996 12:00:00	1/9/1997 12:00:00	1996-12-13 00:00:00	3	7.99	LILA-Supermerc...	Carrera 52 con A...	Barquisimeto	Lara	3508
25	10383	AROUT	8	12/16/1996 12:00:00	1/13/1997 12:00:00	1996-12-18 00:00:00	3	34.24	Around the Horn	Brook Farm Strat...	Colchester	Essex	CO7 6JX
26	10386	FAMIA	9	12/18/1996 12:00:00	1/1/1997 12:00:00	1996-12-25 00:00:00	3	13.99	Familia Arquibal...	Rua Orós 92	Sao Paulo	SP	05442-030
27	10393	SAVEA	1	12/25/1996 12:00:00	1/22/1997 12:00:00	1997-01-03 00:00:00	3	126.56	Save-a-lot Mark...	187 Suffolk Ln.	Boise	ID	83720
28	10394	HUNGC	1	12/25/1996 12:00:00	1/22/1997 12:00:00	1997-01-03 00:00:00	3	30.34	Hungry Coyote l...	City Center Plaza...	Elgin	OR	97827
29	10395	HILAA	6	12/26/1996 12:00:00	1/23/1997 12:00:00	1997-01-03 00:00:00	1	184.41	HILARION-Abast...	Carrera 22 con A...	San Cristóbal	Táchira	5022
30	10398	SAVEA	2	12/30/1996 12:00:00	1/27/1997 12:00:00	1997-01-09 00:00:00	3	89.16	Save-a-lot Mark...	187 Suffolk Ln.	Boise	ID	83720
31	10401	RATTC	1	1/1/1997 12:00:00	1/29/1997 12:00:00	1997-01-10 00:00:00	1	12.51	Rattlesnake Can...	2817 Milton Dr.	Albuquerque	NM	87110
32	10405	LINOD	1	1/6/1997 12:00:00	2/3/1997 12:00:00	1997-01-22 00:00:00	1	34.82	LINO-Delicateses	Ave. 5 de Mayo ...	I. de Margarita	Nueva Esparta	4980
33	10406	QUEEN	7	1/7/1997 12:00:00	2/18/1997 12:00:00	1997-01-13 00:00:00	1	108.04	Queen Cozinha	Alameda dos Ca...	Sao Paulo	SP	05487-020
34	10410	BOTTM	3	1/10/1997 12:00:00	2/7/1997 12:00:00	1997-01-15 00:00:00	3	2.4	Bottom-Dollar ...	23 Tsawassen Bl...	Tsawassen	BC	T2F 8M4

Succeeded (7 sec 28 ms)

Columns 14 Rows 830

Secure and compliant by default

Fabric encryption multi-geo: Data Residency

All data at rest is encrypted by default by Fabric. CMK for OneLake is in limited Public Preview.

Fabric multi-geo capacities allows control over content storage location in one of 54 data centers world-wide.

OneLake which logically spans the world, Workspaces can reside in different regions around the world while still being part of the same data lake.



Compliance with Microsoft Fabric

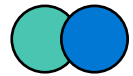
Microsoft Fabric is a core Microsoft online service and supports a wide range of compliance standards including:

- GDPR
- EUDB
- ISO certifications including ISO 27001, 27701, 27017, 27018
- HIPAA, HITRUST compliant
- SOX compliant
- SOC 1, SOC 2, Type 2
- FedRamp HIGH
- PCI DSS

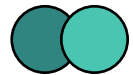


Data security and governance

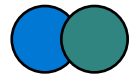
Fortify data security with Microsoft Purview



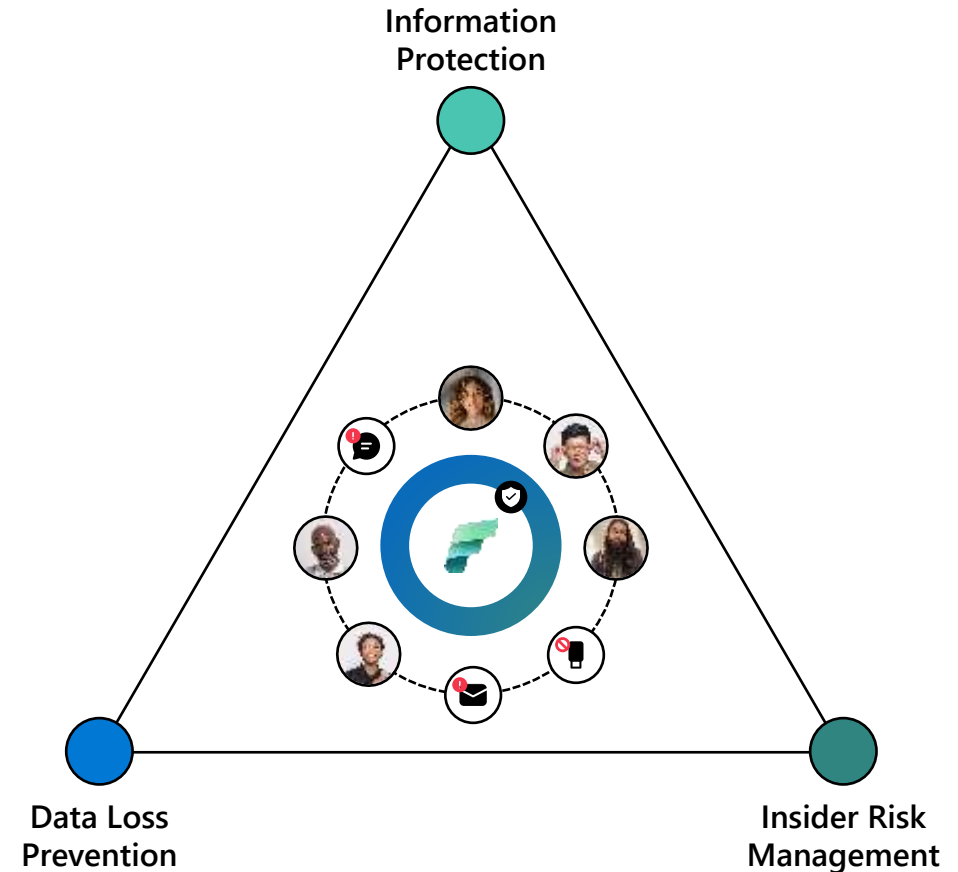
Discover and auto-classify data and prevent it from unauthorized use across apps, services, and devices



Understand the **user intent and context around sensitive data** to identify the most critical risks



Enable **Adaptive Protection** to assign appropriate DLP policies to high-risk users

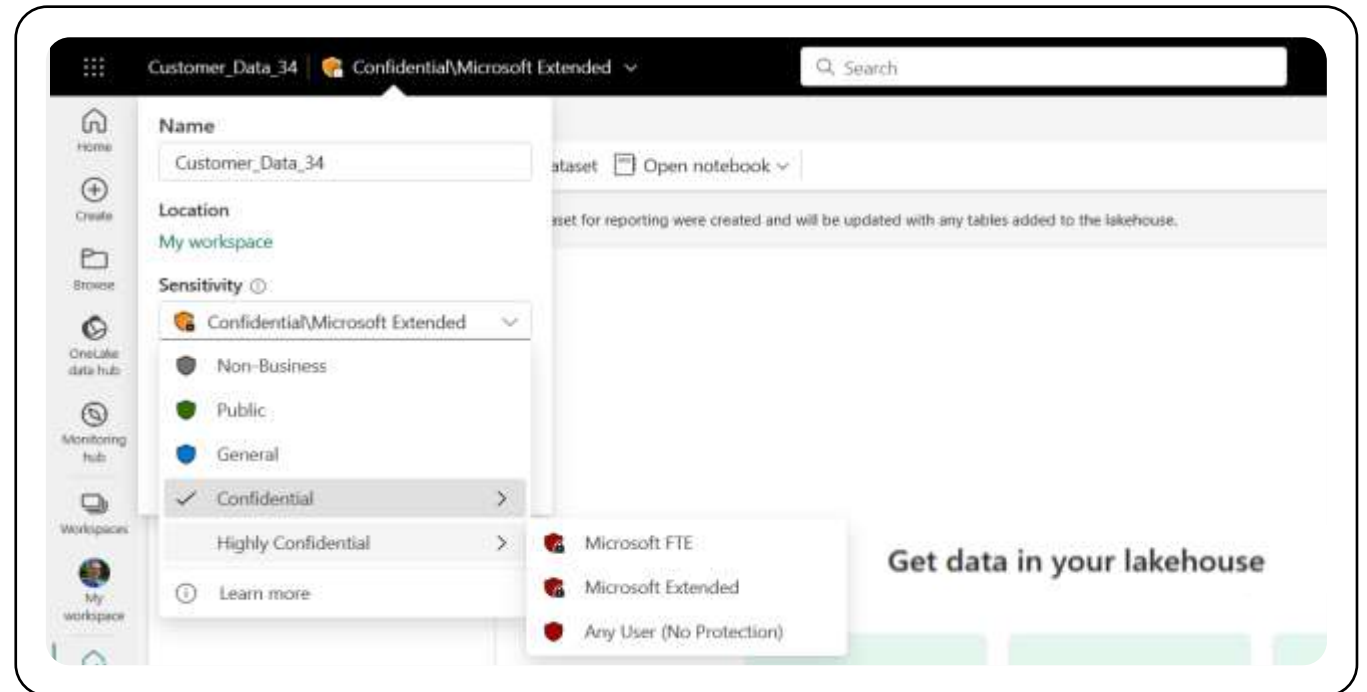


Information Protection labels

Sensitivity Labels **guard sensitive content against unauthorized data access and leakage.**

Travels with the data and enforced even when the data is exported to Office.

Gives **insights into usage** to security admin





Microsoft Purview Information Protection Policies

Data Loss Prevention policies

Define **Purview Data Loss Prevention (DLP) Policies**

Automatically detect sensitive data proactively as it is loaded

Trigger **immediate risk remediation** such as alerts or restrict access.

The screenshot displays the Microsoft Purview Data Loss Prevention (DLP) interface. It features a table of data assets with columns for Name, Type, and Owner. The first asset, 'Human Resources', is highlighted, and its details are shown in a side panel. The details panel includes a description of the lakehouse, a section for policies, and a link to view full details.

Name	Type	Owner
 Human Resources		
 Human Resources		
 Human Resources		
 Customer Satisfaction		
 Dataflow 1		
 Marketing		

Human Resources

This lakehouse contains factors that impact sales and returns. It analyzes which products are most popular in each region and provides.

Policies

Your organization found policy issues in this lakehouse that need your attention.
+ 2 more policies

[See full details](#)



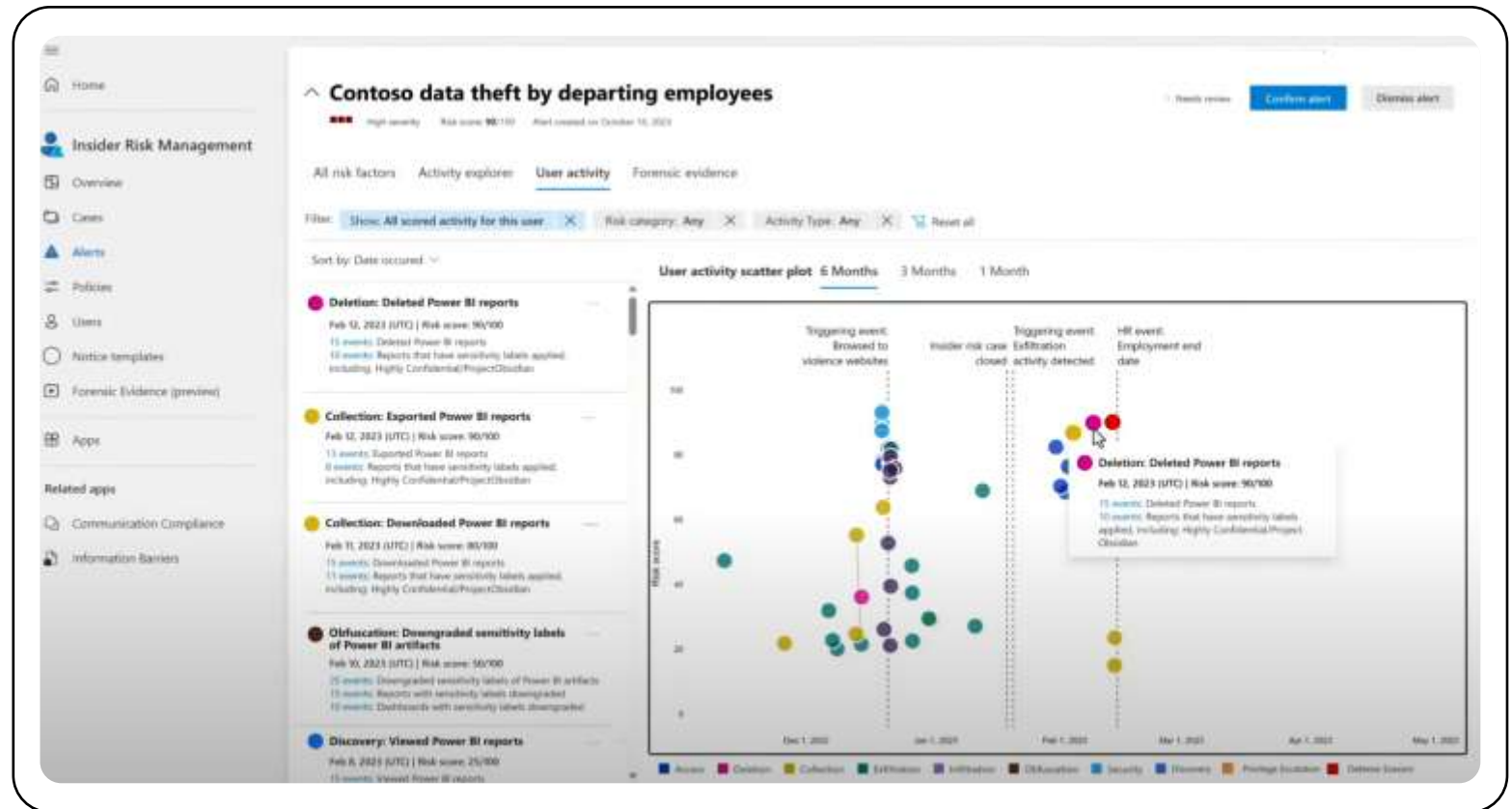
Data Loss Prevention policies for Lakehouse

Insider Risk Management

Detect, **investigate malicious and inadvertent activities** in your organization based on **your organization's compliance standards**.

Also covers **CoPilot** prompts for **Power BI**

Active management of risks through **alerts** and **triage** and **actions**



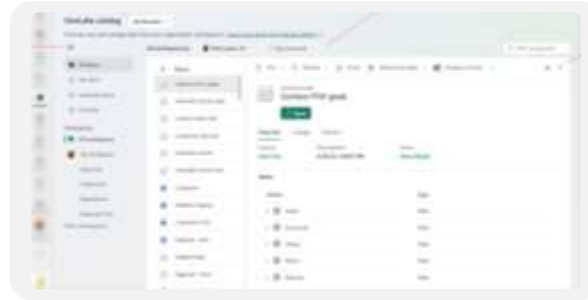
Metadata & Lineage

Explore, manage, and govern data across your organization



Fabric lineage

Track **data lineage** of your analytical projects to **see how data flows** through items and perform impact analysis to **assess impact of changes**. Can be extended with Scanner API's



OneLake catalog

Centrally Find, explore, and use relevant Fabric data items in your organization. Data hub **Pervasive** across Fabric as well as in Teams, Excel and more.



Purview Unified Catalog

Enabling the central data office to **centrally govern and manage data assets**. The Purview Unified Catalog offers data stewards and data owners advanced capabilities for **data curation, advanced data quality, end-to-end data lineage**, and an **intuitive global catalog that spans the data estate**.

Audit

Audit and monitor data across your organization



Admin monitoring & auditing

Uncover insights for effective governance with reports and datasets built for admins. Log, view and export user activities from Microsoft Fabric to support auditing



Purview audit

Microsoft Purview Audit provides with you with the ability to log and search for audited activities and power your forensic, IT, compliance, and legal investigations across your organization.

<https://aks.ms/FabricSecurityWhitepaper>

Thank You !!!